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Insurance Analytics in Healthcare of Elderly People: A ML-based Predictive Model

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Abstract. Effective prediction of insurance costs is a crucial challenge in healthcare economics and insurance analytics. Traditional models often fail to integrate fundamental behavioral and demographic variables that assimilate medical costs, resulting in optimal strategies for cost management. Moreover, existing models have overlooked the cost drives associated with the age variable. In this study, we implement machine learning(ML) models to predict insurance charges by evaluating demographic and behavioral factors. Individuals aged 60 years and older had clear cost-driving trends, indicating that age-based segmentation was given particular attention. The results show that regression analytics and insurance information may increase forecasting accuracy, which can help design insurance policies and hospital resource planning decisions.

Keywords: Length of Stay, Insurance Analytics, Regression, Machine Learning, Hospital Management, Economics, Aging population

1 Introduction

In the modern healthcare system, health insurance plays a critical role in protecting individuals from the rising and often unpredictable costs of medical care. In the U.S., employer-sponsored health insurance premiums in 2023 averaged over 8,400 individuals and 23,900 families [1]. Insurance helps reduce out-of-pocket spending, ensures timely access to care, and improves overall health outcomes [2][3]. It comes in various forms—public, private, and community-based— and aims to spread financial risk and improve accessibility [4] [5].

In recent years, significant digital transformations have been observed in the insurance sector, including AI-driven fraud detection and mobile-based claims processing, leading to greater efficiency and user satisfaction [6] [7]. Understanding how insurance systems evolve is vital for developing sustainable and equitable health care frameworks.

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A growing challenge is the impact of an aging population on the insurance and healthcare systems. Individuals aged 60 years and above contribute to a large portion of healthcare costs due to chronic conditions and longer hospital stays. This demographic shift is expected to drive a sharp increase in healthcare spending in coming decades [17]. While it poses economic concerns, it also creates opportunities for tailored insurance solutions such as risk-adjusted pricing and age-specific preventive strategies.

2 Related Work

Numerous studies have focused on LoS prediction by using electronic medical records, administrative data, and public health datasets. Table 1 summarizes prominent methods in this domain.

Table 1: Summary of State-of-the-Art Methods

Study	Method	Dataset	Remarks
Orji et al. [8]	XGBoost, GBM, RF + XAI	Kaggle	XGBoost: $R^2 = 86.5\%$, best performance
Cenita et al. [9]	LR, GB, SVM	Kaggle	GB: $R^2 = 0.892$, RMSE = 1336.6
Puneeth Kumar et al. [10]	LR + ANOVA	Kaggle	Accuracy $\approx 75\%$; smoking, BMI are key features
Kulkarni et al. [11]	LR, DT, GB	Kaggle	GB: $R^2 = 86.8\%$ (highest among models)
Iqbal et al. [12]	XGBoost, RF, ANN, LR	Kaggle + Custom	Ensemble approach; demographic factors analyzed
Patra et al. [13]	XGBoost, Lasso, Ridge, KNN	Public dataset	XGBoost: $R^2 = 86.8$, RMSE = 4450

3 Methodology

This study uses machine learning models trained on a structured dataset to predict insurance expenses. The methodology consisted of four main stages: data preprocessing, feature correlation analysis, model training, and evaluation. We performed feature selection to find the most significant features by calculating the correlation coefficient between each feature and the "charges" variable.

3.1 Dataset Description

The dataset comprised 1,338 records with seven features: age, sex, BMI, number of children, smoking status, region, and charges. These included numerical

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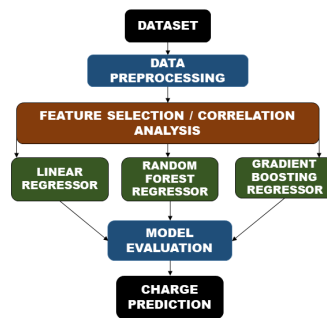


Fig. 1: Workflow of the Proposed Technique

(age, BMI, children, charges) and categorical variables (sex, smoker, region). "Charges" represent the individual's medical cost and are the target variable.

Figure 1 shows the workflow of the proposed technique, where charge prediction is achieved by evaluating models involving a linear regressor, random forest regressor, and gradient boosting regressor after feature selection employing correlation analysis and data preprocessing.

3.2 Data Preprocessing

The dataset used in this study, `insurance_dataset.csv`, was loaded using the `pandas` library in Python. Categorical variables, such as *sex*, *smoker*, and *region*, were transformed into numerical values using `LabelEncoder` to ensure compatibility with machine learning algorithms. Based on preliminary correlation analysis, several features, including *region*, *sex*, *children*, and *BMI*, were removed to minimize potential noise and multicollinearity in the model. The target variable for the prediction was *charges*.

3.3 Model Selection

In this section, we describe the techniques used for regression-based predictions. The proposed approach utilizes an ensemble of machine learning regression models, including linear regression, random forest regression, gradient boosting regression, and XGBoost regression, to enhance prediction accuracy and model robustness. Linear regression assumes a straight-line relationship between variables and serves as a baseline model when the data are linear [18]. A random forest regressor captures nonlinear patterns by averaging multiple decision trees, improving accuracy, and reducing overfitting [19]. The Gradient Boosting Regressor builds models sequentially to correct previous errors using gradient descent,

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thereby effectively handling complex patterns [20]. The XGBoost Regressor enhances gradient boosting with regularization and parallel processing, making it suitable for large-scale high-performance tasks [21].

The models were trained and evaluated to determine the best approach for the regression task. All models were trained using default hyperparameters, with the number of estimators set to 100 where applicable. A fixed random state was used to ensure reproducibility of the results. To facilitate the evaluation, the dataset was split into training and testing subsets using an 80:20 ratio.

3.4 Feature Selection

The proposed model handles a dataset consisting of 1338 samples, with three attributes: Smoker, BMI, and charge. Here, Charge is the dependent variable (target), while Smoker and BMI are independent variables (features). For feature selection, we employed correlation-based feature selection (CFS), which ranks the features based on their correlation with the target variable and their mutual correlation with each other.

The CFS method evaluates the relationship between each feature and the dependent variable (charge), as well as the interrelationships between the features. Features that exhibit a higher correlation with charge are ranked higher, whereas features highly correlated with each other are considered redundant and removed. This process eliminated irrelevant or duplicate features, leading to a more effective and accurate model.

Pearson's correlation coefficient [22] was calculated to assess the linear relationships between the independent variables (Smoker, BMI) and the dependent variable (charge). This calculation allows us to identify which features are significantly related to charge, which may be redundant due to the high correlation between smoking and BMI. Attributes with the highest correlation with charge were selected for further use in the regression model.

Once the correlation matrix was computed and features were selected based on the CFS technique, the final attributes (Smoker and BMI) were used to predict the charge in the regression model.

3.5 Evaluation Strategy

The performance of each regression model was assessed using three commonly used evaluation metrics: the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2 score). MAE quantifies the average magnitude of prediction errors without considering their directions, offering a straightforward interpretation of accuracy [14]. On the other hand, RMSE penalizes larger errors more heavily because of the squaring of residuals, making it particularly useful when large deviations are especially undesirable [15]. The R^2 score measures the proportion of variance in the dependent variable explained by the model, indicating overall model fit and explanatory power [16]. Collectively, these metrics provide a well-rounded evaluation of a model's performance in terms of both accuracy and consistency.

4 Age-Based Segmentation in Insurance Charge Prediction: From Economic Insights to Predictive Application

Several studies, including that of Marešová et al. (2015) [17], have discussed the economic and healthcare challenges of an aging population, although few have explored insurance-specific risk factors in older adults. Individuals aged 60+ face unique health risks that can influence premiums differently from those of the general population. This study focuses on this demographic, analyzing how factors such as smoking and BMI affect insurance charges. Although Marešová et al. [17] highlighted rising costs and non-communicable diseases, they did not address how these risks translate to insurance pricing. To fill this gap, we applied age-based segmentation to individuals aged 60 and above. The findings show that smoking has a stronger link to insurance charges in seniors, whereas BMI is less impactful than in the general population. These results support the need for age-specific models to improve the pricing fairness and accuracy. Building on Marešová et al. [17], this approach integrates demographic segmentation to refine the insurance charge predictions for older adults.

4.1 Understanding the Economic Impact of Aging Populations

Petra Marešová et al.'s (2015) [17] study provides an essential macroeconomic foundation for understanding the implications of population aging. Their research systematically explored how the demographic shift toward an older population, particularly those aged 60 and above, affects healthcare systems and public expenditure. One of their key conclusions is that this age group significantly increases healthcare costs, primarily due to the prevalence of non-communicable diseases (NCDs), such as cardiovascular conditions, diabetes, and respiratory illnesses. Moreover, they emphasize that the long-term economic burden of aging will be placed on both public health systems and individual stakeholders, highlighting the urgent need for proactive policy and financial planning.

4.2 Identifying the Research Gap

However, while Marešová et al. [17] offer crucial insights into the economic and systemic effects of an aging society, their work does not explore how these macro trends translate into individual-level insurance risk predictions. They stop short of modeling how specific risk factors, such as smoking and BMI, contribute to cost variability in older populations. This creates an important research gap, especially in the insurance industry, where individual-level pricing and risk stratification are essential for sustainable policy development and fairness.

4.3 Operationalizing Economic Trends into Predictive Models

This study addresses this gap by operationalizing the macro-level implications of Marešová et al. [17] into a micro-level predictive framework. Specifically, by

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segmenting the population dataset to focus only on individuals aged 60 years and above, this study examines how key health-related behaviors (smoking and BMI) interact with insurance charges in this distinct demographic group. Segmentation not only reveals differential patterns of risk, but also allows for the development of age-specific models that better align with the lived realities of older adults.

4.4 Synthesis and Implications of Combined Approaches

Marešová et al.[17] identified that aging leads to greater healthcare expenditures and economic stress, showing how and why such financial burdens materialize differently for senior individuals within predictive insurance contexts. The stronger correlation between smoking and insurance charges for older adults found in this analysis reinforces the confounding risk effects of such behavior over time. Meanwhile, the weaker association between BMI and charges in seniors challenges the assumptions often made in broader predictive models and suggests that weight-based risk may need to be reconsidered in the context of age.

Together, the two works form a complementary framework: Marešová's study provides a strategic and economic rationale for focusing on seniors, whereas this study delivers a technical and applied approach to accurately model and manage that risk. By bridging macroeconomic insight with machine learning-driven segmentation, the combined approach not only advances academic understanding, but also holds practical value for insurers and policymakers aiming to build more equitable and sustainable models in an aging world.

In essence, where Marešová's work highlights why the aging population matters economically, this study demonstrates how that importance can be quantified and modeled in insurance terms, ultimately enabling more nuanced data-driven decisions in both public health and private sector risk management.

5 Experimental Results and Discussion

In this section, we describe the feature selection techniques, ablation studies, and performance of different regressors using a combination of the best features.

5.1 Feature Selection

Two techniques were used for feature selection. The first is based on correlation calculation between each input feature and the target variable ("charge"), and The second is feature selection using Random Forest Regressor, which ranks features based on their importance in reducing prediction error across multiple decision trees.

The correlation heatmap presented in Figure 2 depicts the linear relationship between various independent features and the target variable, insurance charges. This analysis was conducted using Pearson's correlation coefficients [22] to determine the strength and direction of the associations between predictors and outcome variables.

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As observed, the "smoker" feature demonstrates the strongest positive correlation with insurance charges (correlation coefficient = 0.79), indicating that smoking status is a highly influential factor in determining medical costs. "Age" also exhibits a moderate positive correlation (0.30), suggesting that "insurance charges" increase with advancing age. In contrast, "BMI" showed a weaker positive correlation (0.20), indicating a relatively minor impact. Other variables such as "children" (0.07), "sex" (0.06), and "region" (-0.01) exhibit negligible correlation, implying limited direct influence on the target variable.

These results highlight the significance of smoker and age as primary predictors of insurance charges, consistent with domain knowledge. This correlation analysis provides a valuable foundation for feature selection in model development and supports the interpretability of the proposed predictive framework.

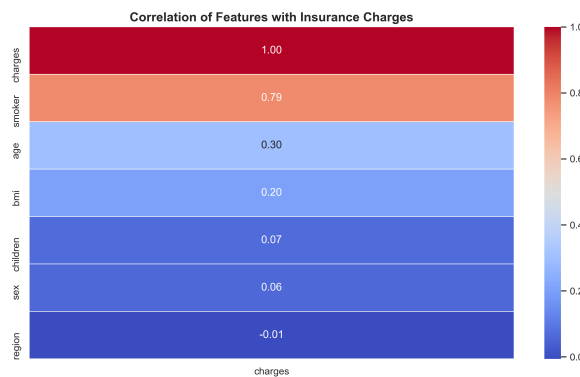


Fig. 2: Feature-Target Correlation Heatmap

However, feature selection using Random Forest Regressor shown in Figure 3 indicates "smoker" and "BMI" to be the most important features, followed by "age", "children," "region," and "sex."

Therefore, we conducted an elaborate ablation study to verify which features are important for charge prediction, as shown in Table 2.

5.2 Ablation Study

Table 2 shows the performance of the gradient boosting regressor for charge prediction using combinations of different input features.

5.3 Model Performance

After the ablation study reported in Table 3, we concluded that the combination of the following features—age, smoking, sex, BMI, and children—had the highest impact on charge prediction. We used three main regression models for

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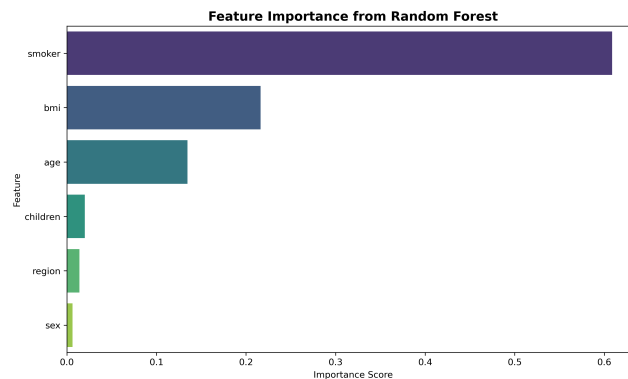


Fig. 3: Feature Importance from Random Forest Regressor

Table 2: Ablation study showing how different feature combinations affect RMSE, MAE, and R^2 scores for predicting medical charges. The boldface font shows the combination that has the highest performance metric.

Experiment	Age	Sex	BMI	Children	Smoker	Region	RMSE / MAE / R^2
Only Age	✓						11834.27 / 9175.58 / 0.3016
Only Smoker					✓		7262.67 / 5625.85 / 0.7902
Only Sex		✓					12443.60 / 9542.20 / 0.0626
Only BMI			✓				13028.53 / 10173.23 / 0.2017
Only Children				✓			12419.43 / 9577.61 / 0.0065
Only Region						✓	12394.19 / 9554.82 / -0.0105
Age + Smoker	✓				✓		6270.31 / 4035.89 / 0.7467
Smoker + BMI	✓		✓		✓		5878.96 / 4470.05 / 0.7774
Age + Smoker + Sex	✓	✓			✓		6304.58 / 4032.95 / 0.7440
Age + Smoker + Sex + BMI	✓	✓	✓		✓		4502.55 / 2545.43 / 0.8694
Age + Smoker + Sex + BMI + Children	✓	✓	✓	✓	✓		4342.91 / 2424.53 / 0.8785
All Features	✓	✓	✓	✓	✓	✓	4352.54 / 2447.95 / 0.8780

performance evaluation: linear regressor, random forest regressor, and gradient boosting regressor.

Table 3: Performance of different regressor models for charge prediction using a combination of age, smoker, sex, BMI, and children features.

Model	MAE	RMSE	R^2 Score
Linear Regressor	4213.48	5829.17	0.7811
Random Forest Regressor	2549.01	4703.25	0.8575
Gradient Boosting Regressor	2424.53	4342.91	0.8785

From Table 3, we can see that among all the regression models, the gradient boosting regressor has the highest performance and the Linear Regressor has the lowest performance. Furthermore, in Figure 4, we plot the actual versus

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predicted insurance charges using the Gradient Boosting, Random Forest, and Linear Regressor in (a), (b), and (c), respectively.

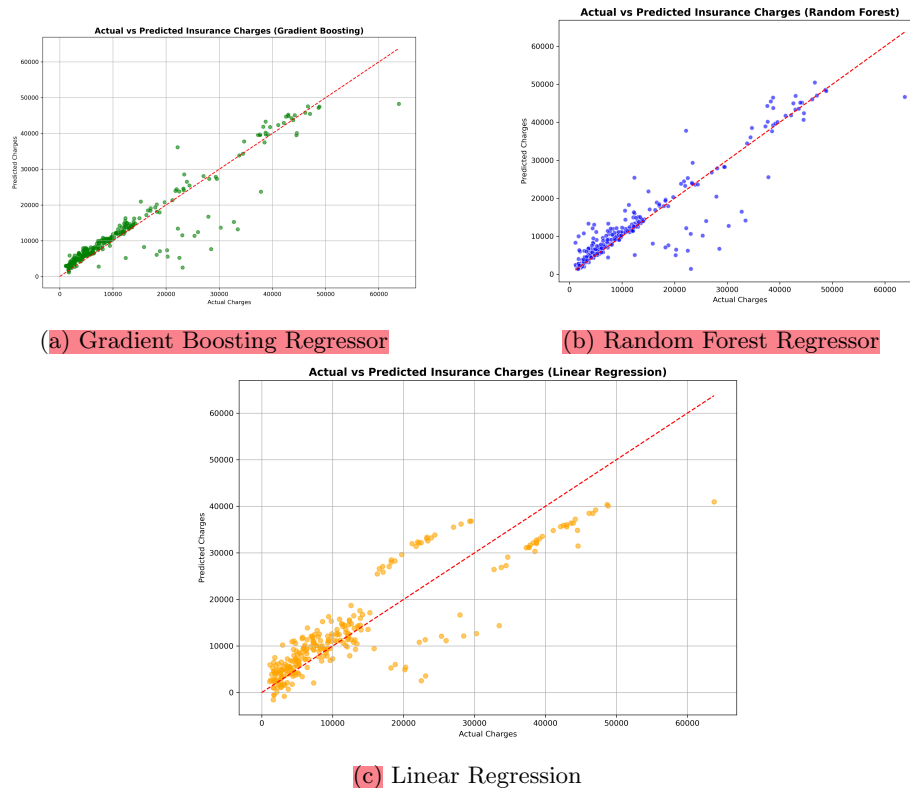


Fig. 4: Actual vs Predicted Insurance Charges using different regressors: (a) Gradient Boosting, (b) Random Forest, and (c) Linear Regression.

5.4 Insights

The ensemble-based models outperformed linear regression-based Gradient Boosting. Boosting provides the most robust and accurate results. These findings highlight the importance of using advanced regression techniques in insurance-based business analytics.

5.5 Integration of Age-Based Analysis

Several studies, including Marešová et al. (2015) [17], have discussed the economic and healthcare challenges of an ageing population, though few explore

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insurance-specific risk factors in older adults. Individuals aged 60+ face unique health risks that can influence premiums differently from the general population. This study focuses on that demographic, analyzing how factors like smoking and BMI affect insurance charges. While Marešová et al. [17] highlight rising costs and non-communicable diseases, they do not address how these risks translate to insurance pricing. To fill this gap, we applied age-based segmentation for individuals 60 and above. Findings show smoking has a stronger link to insurance charges in seniors, whereas BMI is less impactful than in the general population. These results support the need for age-specific models to improve pricing fairness and accuracy. Building on Marešová et al. [17], this approach integrates demographic segmentation to refine insurance charge predictions for older adults.

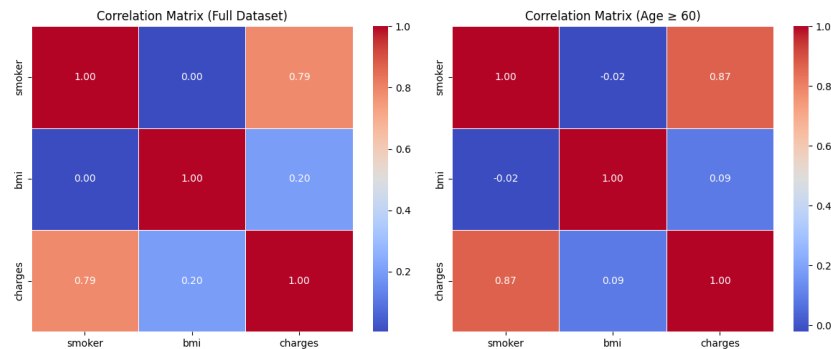


Fig. 5: Correlation Matrix for Full Dataset and Age 60+

Smoking and Healthcare Charges Correlation . As shown in Figure 5, the correlation between smoking factors and healthcare charges in the full dataset is 0.787, indicating a strong positive relationship. In particular, this correlation increases to 0.869 in individuals aged 60 years and above, suggesting that smoking has a significantly greater impact on healthcare costs in older adults. This reflects the compounded health risks and long-term medical needs associated with smoking in this demographic group. These findings underscore the importance of targeted smoking cessation programs in the elderly population.

BMI and Healthcare Charges Correlation According to Figure 5, the correlation between BMI and healthcare charges was 0.198 in the full dataset, implying a weak relationship. For individuals aged 60 years and above, this correlation further declined to 0.088, indicating that BMI plays a minimal role in predicting healthcare costs among seniors. This suggests that other factors, such as smoking and chronic health conditions, may have a stronger influence on healthcare expenditure in older adults.

Smoking and BMI Correlation Figure 5 also shows that the correlation between smoking status and BMI was negligible in both the full dataset (0.0037) and the 60+ age group (−0.0218). This indicates no meaningful relationship between the two variables, suggesting that public health interventions targeting smoking and weight management should be approached independently, particularly for older individuals.

6 Conclusion and Future Work

This study introduced a machine-learning-based approach using demographic and behavioral attributes to predict hospital length of stay and associated insurance charges. Feature selection and ablation analysis identified smoking status, age, and BMI as key predictors, particularly in older adults. Ensemble regression models, notably Gradient Boosting, have demonstrated superior performance over traditional linear approaches, underscoring their suitability for healthcare cost prediction. By incorporating age-based segmentation, the model provides tailored insights for high-risk groups, offering practical implications for both insurers and hospital administrators in planning and risk assessment. Future research will aim to integrate the model into real-time electronic health record (EHR) systems, incorporate clinical features, such as comorbidities and laboratory values, and evaluate the framework across diverse geographic and institutional datasets. Additionally, the development of interactive dashboards may support real-time decision-making in hospital operations.

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